

EE366200 DSP Laboratory

Lab 7 Image Filtering and Corner Detection

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1. Introduction

This lab investigates interest-point detection in images through the Harris corner detector and demonstrates a simple application of those detected points to infer geometric shift between two frames. The work is framed around the classical second-moment formulation in which local intensity variations are summarized by a 2×2 matrix computed from image gradients within a neighborhood; the eigen-structure of this matrix characterizes whether a location behaves like a flat region, an edge, or a corner. Building on that model, the lab defines the Harris response function to score each pixel's "corneriness" and then extracts a sparse, well-localized set of salient points by retaining only strong, isolated maxima. The provided materials pair this theory with runnable code that converts images to grayscale, computes horizontal and vertical derivatives, aggregates gradient products over a weighted window to form the second-moment terms, evaluates the Harris response, and selects final detections through thresholding and non-maximum suppression. A companion script applies the same detector to two images of a scene containing a rectangle, overlays the recovered points for visualization, and uses them to estimate the rectangle's displacement between the

frames. In this way, the lab connects the mathematical idea of second-order intensity structure to a practical, end-to-end pipeline for detecting corners and using them for a basic motion-measurement task, while keeping the focus on how the windowed gradient statistics and the Harris response together yield stable and repeatable points for subsequent analysis.

2. Objectives

This lab has two end goals. The first is to demonstrate corner detection on a natural image by running a Harris-based detector and presenting the resulting evidence in figures: the original frame, the Gaussian-smoothed cross-gradient map I_{xy} , and the final overlay of detected interest points, along with the reported count of corners. The second is to quantify the translation of a synthetic square between two frames by detecting its four vertices in each image, visualizing the results in figures analogous to the first task, and reporting the measured inter-frame displacement as row and column shifts.

3. Approach

The method starts by converting each RGB image to a single luminance channel so that all subsequent operations act on comparable intensity values. Spatial derivatives along the horizontal and vertical directions are then obtained by convolving the grayscale image with a central-difference kernel, yielding I_x and I_y . From these

derivatives the three gradient-product images I_x^2 , I_y^2 , and $I_x I_y$ are formed and smoothed with a Gaussian window to aggregate neighborhood information while suppressing noise. At every pixel, the smoothed products define the entries of the second-moment matrix:

$$A = \sum_{x,y} w(x,y) \begin{pmatrix} I_x^2 & I_x I_y \\ I_x I_y & I_y^2 \end{pmatrix}$$

and the Harris response is evaluated as $R = \det(A) - k(\text{trace } A)^2$. The response image is normalized to a convenient numeric range and pruned into a sparse, well-localized set of interest points by applying a threshold followed by non-maximum suppression in a $(2r + 1) \times (2r + 1)$ neighborhood. Detections too close to the image border are discarded to avoid windows that fall outside the frame. The same detector, together with identical derivative filters and Gaussian parameters, is reused across all experiments so that differences in the outputs arise from image content rather than algorithmic changes.

After the procedure above is established, two tasks demonstrate their behavior and outputs. The first task is a corner-detection run on a natural scene. Figure 1 shows the original house image used as input. Figure 2 visualizes the Gaussian-smoothed cross term I_{xy} ; the bright and dark ridges along the roofline indicate strong, coherent changes in orthogonal gradient directions, which foreshadow high Harris responses near roof joints and window grills. Figure 3 overlays the final detections on the image after

thresholding and non-maximum suppression. In this run the script reports 106 corners, which cluster around architectural junctions such as roof peaks, balcony intersections, window frames, and step edges—precisely the places where intensity structure varies significantly in both directions.



Fig.1 The original house image

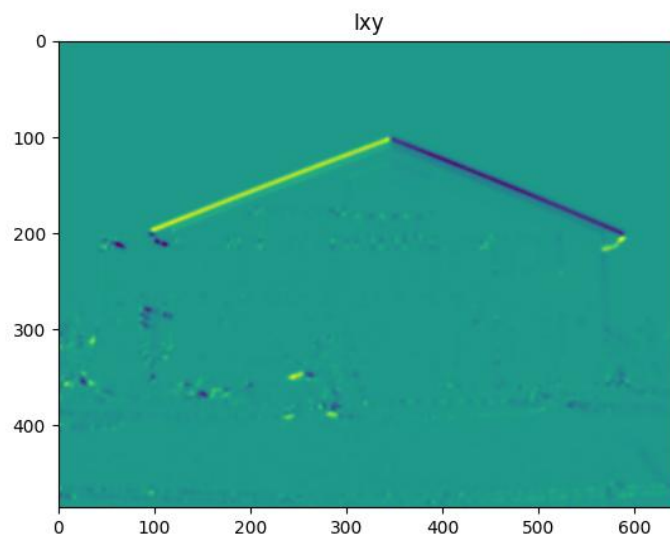


Fig.2 Visualization of I_{xy} of the house image



Fig.3 Detected corners of the house image

The second task measures the translation of a synthetic square between two frames. Figure 4 is the first frame, where the bright box appears in the upper-left region; Figure 5 shows its I_{xy} map with four compact extrema near the vertices, reflecting cross-directional changes at the corners. Figure 6 is the second frame, in which the same box has moved toward the lower-right; Figure 7 gives the corresponding I_{xy} map for this frame. The detector outputs for the two frames are visualized in Figure 8 and Figure 9, each displaying four red points placed precisely at the box vertices. By comparing corresponding vertex coordinates across the two detections, the script computes a row shift of 160 pixels and a column shift of 80 pixels. The magnitude and direction of this estimate are consistent with the visual displacement apparent between the two “Original Image” figures, confirming that the same corner-selection pipeline used for the natural

scene can also support a straightforward motion-measurement task when the scene undergoes a pure translation.

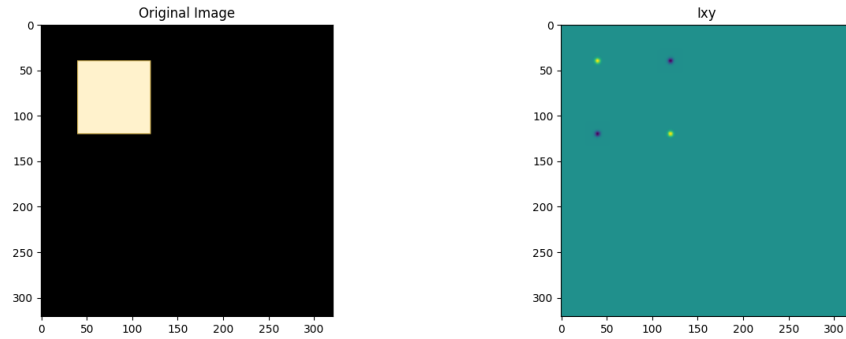


Fig.4 The first box image (left), Fig.5 Visualization of I_{xy} of the first box image (right)

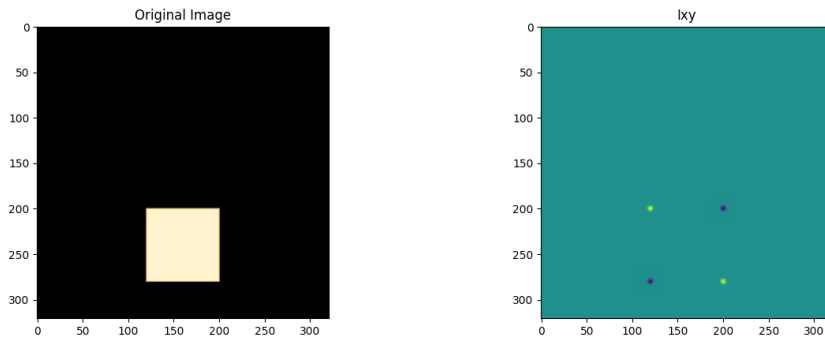


Fig.6 The second box image (left), Fig.7 Visualization of I_{xy} of the second box image (right)



Fig.8 Detected corners of the first box (left), Fig.9 Detected corners of the second box (right)

4. Discussion

- I. Show at least two different corner image generation results and each needs to contain the input image, Gaussian smoothed I_{xy} and corner detection result.

Ans:

Fig.10–Fig.15 present two additional corner-detection examples. Fig.10 and Fig.13 show the original inputs; Fig.11 and Fig.14 display the corresponding Gaussian-smoothed cross-gradient maps I_{xy} ; and Fig.12 and Fig.15 overlay the detected corners. In these experiments, the detector finds 108 corners in the first image and 35 in the second.

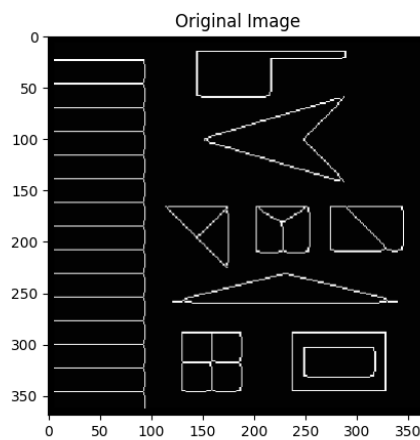


Fig.10 The first original image

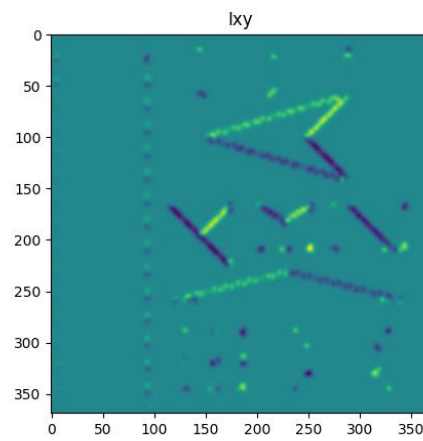


Fig.11 Visualization of I_{xy} of the first image

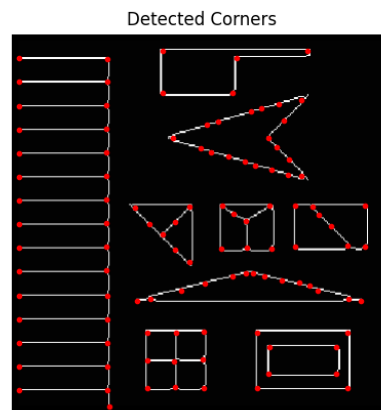


Fig.12 Detected corners of the first image

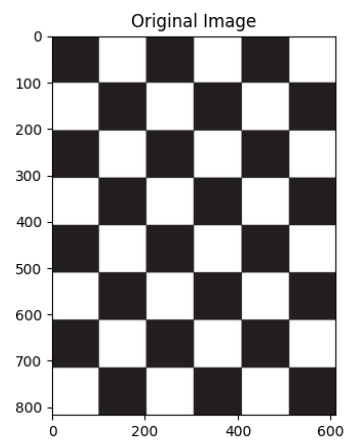


Fig.13 The second original image

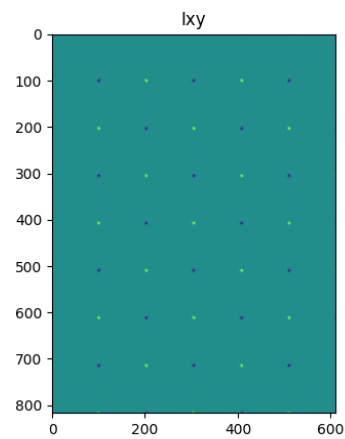


Fig.14 Visualization of I_{xy} of the second image

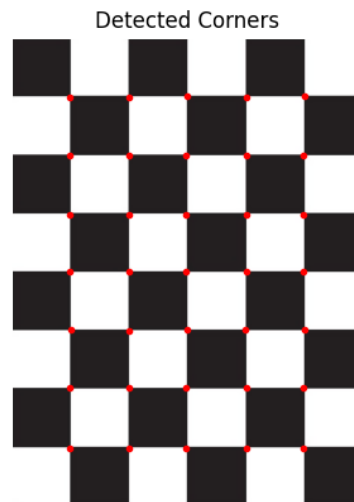


Fig.15 Detected corners of the second image

II. Why using Gaussian window to get Gaussian smoothed I_x, I_y, I_{xy} ?

Ans:

The gradient images are locally integrated to form the second-moment (structure-tensor) terms, and the choice of window controls how that neighborhood evidence is pooled. A Gaussian window provides an isotropic,

smoothly decaying weighting around each pixel, which stabilizes the tensor against noise while retaining locality. In standard treatments of the structure tensor, this integration is explicitly performed with a Gaussian so that gradient covariances are estimated from a neighborhood without introducing directional bias. This produces reliable eigenvalues and eigenvectors for corner/edge classification and is the default formulation in modern references.

Beyond empirical robustness, the Gaussian window is theoretically singled out by linear scale-space theory. If the smoothing stage is required to be linear and shift-invariant and to relate fine and coarse representations without creating new structures, then the Gaussian is the unique kernel that satisfies these axioms. Using it to integrate gradient products therefore embeds the Harris operator in a well-behaved multi-scale framework where the scale parameter σ directly controls the size of the neighborhood contributing to the tensor.

Operational benefits follow from these properties. The Gaussian is separable and closed under convolution, enabling efficient implementation and consistent behavior across scales; repeated small Gaussian smooths are equivalent to one larger smooth, which makes it convenient to tune or compound integration scales in practice.

Finally, using a Gaussian window on I_x^2 , I_y^2 , and $I_x I_y$ reduces spurious extrema in the response while preserving true corner evidence. Analytical and experimental studies of the Harris detector model its autocorrelation (second-moment) matrix explicitly as gradient terms convolved with a Gaussian, and evaluate how different gradient masks and Gaussian integrations affect precision and runtime, reinforcing that Gaussian weighting is integral to the detector's stability.

In summary, Gaussian windowing supplies isotropic, noise-suppressing neighborhood integration; it places the detector on firm scale-space foundations; it affords efficient and consistent multi-scale computation; and it yields stable corner responses—all of which are central to forming reliable I_x , I_y , and I_{xy} inputs to the Harris second-moment matrix.

III. Why do we need the threshold in procedure VI?

Ans:

After the Harris response R is computed, the result is a dense scalar field in which true corner evidence coexists with low-amplitude fluctuations from noise, weak textures, and numerical effects. A threshold in Procedure VI imposes a minimum strength requirement so that only pixels with sufficient “cornerness” are admitted as candidates. In this lab, R is first normalized so that the maximum

equals a fixed value (set to 1000 in the code), and a constant threshold (e.g., 20) is then applied. This normalization makes the cutoff meaningful and comparable across images, while the threshold itself converts the continuous response into a discrete set of plausible features.

The threshold is necessary even when non-maximum suppression is used. Non-maximum suppression guarantees spatial distinctness by keeping only local peaks, but it does not distinguish between a strong, structurally meaningful peak and a small ripple created by noise. The threshold removes broad regions whose R values remain uniformly small, preventing weak peaks from entering the candidate set. This two-step process—apply a response floor, then thin spatially—produces a sparse and reliable set of detections that can be overlaid on the image and used in subsequent tasks.

The choice of threshold directly controls feature density and stability. In the house example, using the fixed normalization and the specified cutoff yields 106 detected corners, which cluster at roof joints, window frames, and other high-contrast junctions. Lowering the threshold would admit many additional points from textured surfaces and edges, increasing false positives; raising it would retain only the most prominent junctions, possibly missing useful features. The same mechanism underpins the rectangle experiment: with the threshold in place, only

the four vertex responses exceed the cutoff in each frame, so the detector returns precisely four points per image. Those stable detections then support a clean shift estimate of 160 pixels in rows and 80 pixels in columns when corresponding vertices are compared between frames.

Practically, the threshold also interacts with the non-maximum suppression radius and the Gaussian integration scale. A larger radius suppresses nearby peaks so that the remaining corners are more widely separated, while the threshold determines whether a surviving peak is accepted at all. A larger Gaussian scale increases spatial averaging in the second-moment terms and typically raises strong corners while damping small fluctuations; the threshold ensures that only peaks that remain significant after this smoothing are kept. Together, these elements align with the lab's procedure and code: normalize R , apply a fixed threshold, perform non-maximum suppression, and mask borders, yielding a compact, reproducible set of corners suitable for visualization and for downstream measurements such as the box translation.

- IV.** Analyze the result of using Rectangular window function and try to discuss the difference between Rectangular window function and Gaussian window function. Besides, please try your own window function and explain the reason why you choose it. Rectangular window function (use the same size as the provided

Gaussian filter)

Ans:

Using the same kernel size for the integration window, the rectangular window yields 64 detected corners on the house image (see Fig.16 for the I_{xy} map and Fig.17 for the overlay), while the Hanning window yields 60 detected corners (Fig.18–Fig.19). The rectangular I_{xy} visualization in Fig.16 exhibits broader, block-like high responses along the roof edges and scattered patches in relatively flat regions; the overlay in Fig.17 correspondingly includes several peripheral points and edge-adjacent detections. By contrast, the Hanning results are smoother in Fig.18 and concentrate near true geometric junctions; Fig.19 shows fewer isolated detections away from structure.

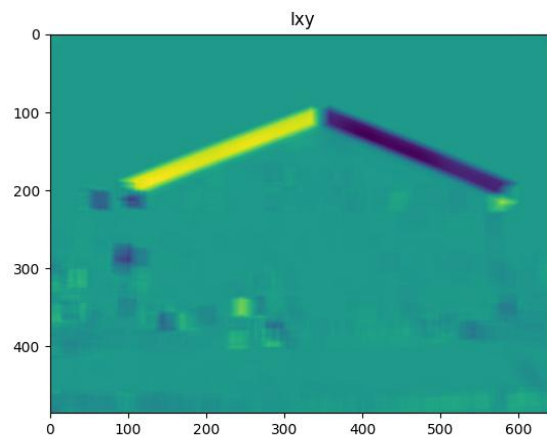


Fig.16 Visualization of I_{xy} of the house image using rectangular window



Fig.17 Detected corners of the house image using rectangular window

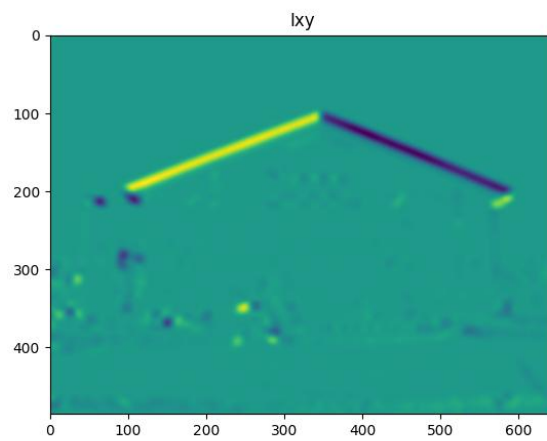


Fig.18 Visualization of I_{xy} of the house image using Hanning window



Fig.19 Detected corners of the house image using Hanning window

An additional observation from Fig.17 is that many of the rectangular-window detections are not precisely located at the true corner pixels. Instead, the markers tend to sit a few pixels away from the ideal roof apex or window-frame intersection. This slight offset is consistent with the rectangular window's hard edges: equal weighting throughout the footprint and an abrupt cutoff at the boundary allow strong edge energy to "leak" into neighboring positions, which can bias the second-moment estimate and shift the Harris peak off the exact junction. Tapered windows (Gaussian/Hann) reduce this effect by down-weighting distant samples smoothly, thereby improving localization at the actual intersection points; this is reflected by the tighter alignment of detections with roof and frame corners in Fig.19. The underlying trade-off—rectangular windows have narrow main lobes but high sidelobes, whereas tapered windows widen the main lobe and suppress sidelobes—is well established in window analysis and explains the visual differences seen here.

Relative to the earlier Gaussian run on the same image (106 corners), both rectangular and Hanning produce smaller, more selective sets under the same threshold and NMS settings. The Gaussian's smoothly decaying weights emphasize the central neighborhood and typically yield richer yet still sensible coverage of salient junctions, while the rectangular window's equal weighting

increases susceptibility to edge leakage and localization bias, and the Hanning taper strikes a middle ground by suppressing sidelobes with finite support. These qualitative behaviors match standard characterizations of the three windows (rectangular: highest sidelobes; Hann/Hanning: lowered sidelobes; Gaussian: smooth, scale-space consistent weighting).

Based on the visual evidence alone, selecting the Hanning window as the additional choice is reasonable. Its ($I_{\{xy\}}$) map exhibits fewer broad background undulations, and the final corners are concentrated at geometric junctions—such as the roof ridge vertex, balcony corner, and window-frame intersection. Although the corner count (60) is slightly lower than with the Rectangular window (64), the detections are visually cleaner, with fewer scattered points away from structure, and they show better localization at the true junctions. Under this lab's fixed threshold and NMS settings, Hanning strikes a more balanced trade-off between suppressing non-structural responses and retaining genuine corners, while the Gaussian window remains the baseline that produced the richest yet still sensible distribution of corners on this image. Notably, in Fig.17 many Rectangular detections lie a few pixels off the actual corner locations, whereas the Hanning (Fig.19) and Gaussian results align more precisely with the true intersections.

- V. Analyze the result of using Sobel and Scharr gradient filter. Besides, please try to discuss the difference between Sobel and Scharr gradient filter. Below shows these two gradient filters.

Ans:

Using the house image and the same detector settings, the two Sobel results (Fig.20–Fig.21) and the two Scharr results (Fig.22–Fig.23) produce very similar I_{xy} visualizations and nearly identical corner overlays. In fact, the corner count is 106 for all three gradient choices in this lab (the original central-difference baseline, Sobel, and Scharr). From the provided figures alone, any difference is minimal: the major junctions (roof apex, balcony tips, window-frame intersections) are recovered in all cases, and the spatial layout of detections remains effectively the same.

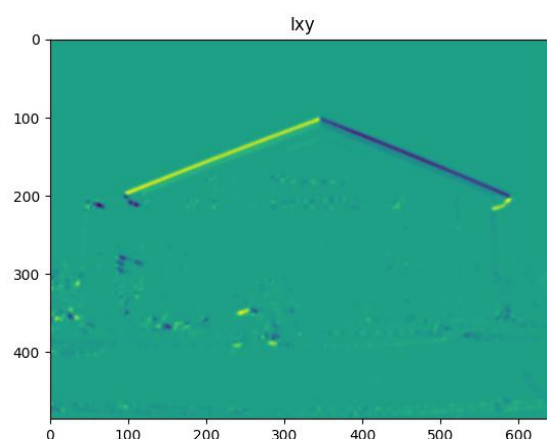


Fig.20 Visualization of I_{xy} of the house image using Sobel gradient filter



Fig.21 Detected corners of the house image using Sobel gradient filter

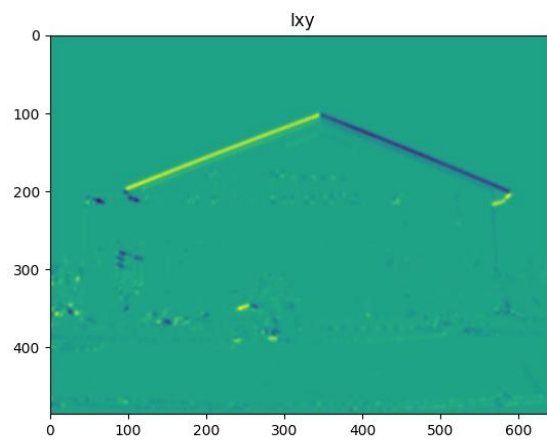


Fig.22 Visualization of I_{xy} of the house image using Scharr gradient filter



Fig.23 Detected corners of the house image using Scharr gradient filter

The distinction between the operators is chiefly theoretical. Sobel is a discrete differentiation operator built from a central-difference in the derivative direction combined with a short smoothing in the perpendicular direction, implemented by separable 3×3 integer kernels; it is inexpensive and provides a noise-reducing gradient estimate for I_x and I_y (and hence I_{xy}). Scharr belongs to the same family of small-kernel derivative operators but uses coefficients that are optimized for better rotational symmetry (isotropy) and lower angular error when estimating gradient orientation with a 3×3 mask; the commonly used 3×3 Scharr pair and its transpose arise from an optimization that minimizes a weighted mean-squared angular error in the Fourier domain. Scharr is therefore expected to respond more uniformly to edges at different orientations and to give slightly sharper gradients in theory, even though, in the images provided here, the downstream Harris corners are effectively unchanged. These observations also align with the structure-tensor stage of Harris: once gradients are aggregated over a Gaussian window, modest differences in the raw derivative masks can be attenuated, which helps explain why Sobel and Scharr lead to the same 106-corner outcome in this dataset.

5. Conclusion

This lab implemented a Harris-based corner detection pipeline and verified it end to end on both a natural scene and a synthetic translation case. The detector consistently recovered salient architectural junctions on the house image and measured the box motion between two frames as a 160-pixel row shift and an 80-pixel column shift, in line with the expected behavior of the second-moment formulation that underpins the Harris response.

Across window choices, the Gaussian integration served as a strong baseline, yielding 106 well-distributed corners on the house image. The rectangular window, used at the same footprint, produced 64 corners and showed broader I_{xy} bands with detections that were sometimes a few pixels off the true junctions, reflecting the hard cut-off of equal weighting. The Hanning window gave 60 corners with cleaner maps and tighter alignment to roof and window intersections, illustrating the practical value of smoothly decaying weights for the structure-tensor estimate. These observations are consistent with the general principle that smoothly weighted neighborhoods stabilize the second-moment matrix, which is the basis for the Harris measure.

With respect to gradient operators, replacing the central-difference baseline with Sobel or Scharr did not materially change outcomes on this dataset: all three variants produced the same total of 106 corners and visually similar overlays. Theoretical

differences remain—Sobel mixes a short smoothing with a central difference, whereas Scharr adjusts 3×3 coefficients to improve isotropy and angular accuracy—but the subsequent Gaussian integration in the Harris pipeline attenuated these small gradient-level differences, leading to near-identical detections.

Overall, the results demonstrate that the Harris detector, coupled with Gaussian neighborhood integration, provides stable and repeatable corners suitable for basic motion measurement, while tapered windows such as Hanning can reduce peripheral or mis-localized responses without sacrificing key features. Future work could explore multi-scale settings and alternative response functions within the same second-moment framework to test robustness under larger appearance changes.

6. Reference

- [1] Department of Electrical Engineering, National Tsing Hua University, *EE3662-2025-Unit II-Visual Signal*.
- [2] Department of Electrical Engineering, National Tsing Hua University, *Lab7*.